

Melatonin and sleep: Exploring its role in regulating the circadian rhythm and sleep-wake cycle

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ABSTRACT

Sleep is a vital physiological process essential for maintaining overall health, supporting cellular repair, immune regulation, and cognitive functions. Sleep disruptions are associated with adverse health outcomes, including cardiovascular diseases, metabolic disorders, and impaired mental health. Sleep regulation is orchestrated by the circadian rhythm and homeostatic sleep drive, with melatonin—a neurohormone secreted by the pineal gland—playing a pivotal role in modulating these processes. Melatonin synthesis follows a biochemical pathway, converting serotonin to melatonin through key enzymes, AANAT and HIOMT, under circadian control. The suprachiasmatic nucleus (SCN) governs its secretion via light-responsive pathways, with light exposure inhibiting and darkness stimulating melatonin production. Acting through MT1 and MT2 receptors in the SCN and other brain regions, melatonin facilitates sleep onset, circadian rhythm alignment, and sleep maintenance while exerting antioxidant, anti-inflammatory, and chronobiotic effects. Exogenous melatonin has emerged as a therapeutic agent for managing sleep disorders, such as circadian rhythm sleep-wake disorders, jet lag, and insomnia. Its chronobiotic properties enable circadian phase-shifting, aligning sleep-wake cycles with environmental cues. Despite its growing use, questions remain regarding optimal dosing and long-term safety. This review delves into the physiology of melatonin, its synthesis, regulation, and mechanisms of action in sleep regulation. It also examines its therapeutic potential, highlighting gaps in research and paving the way for future studies on melatonin's broader implications in health and disease.

Keywords: Antioxidant, chronobiotic, circadian rhythm, exogenous melatonin, insomnia, jet lag, light exposure, melatonin, MT1/MT2 receptors, pineal gland, SCN, sleep disorders, sleep, sleep-wake cycle

Brief Overview of the Importance of Sleep in Health

Sleep is a fundamental physiological process essential for maintaining overall health and well-being. It is a complex and dynamic state that supports numerous biological functions, including cellular repair, immune system regulation, and cognitive processing. Adequate sleep is vital for optimal physical, mental, and emotional health. Chronic sleep deprivation or disturbances

have been linked to a wide range of adverse health outcomes, including cardiovascular diseases, metabolic disorders such as diabetes, impaired immune function, and mental health conditions like depression and anxiety. Moreover, sleep plays a pivotal role in memory consolidation and learning, underlining its importance for cognitive performance.^[1]

Sleep is regulated by two main processes: the circadian rhythm and the homeostatic sleep drive. The circadian rhythm is governed by the body's internal clock, located in the hypothalamus's suprachiasmatic nucleus (SCN), which synchronizes physiological processes to the 24-hour light-dark cycle. The homeostatic drive ensures that the longer an individual stays awake, the greater the pressure to sleep. Disruptions to these processes, caused by

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factors such as shift work, jet lag, or excessive screen time, can lead to sleep disorders and negatively impact health outcomes.^[2]

Introduction to Melatonin as a Critical Neurohormone Involved in Sleep Regulation

Melatonin, often called the “hormone of darkness,” is a neurohormone primarily secreted by the pineal gland in response to darkness. It plays a central role in the regulation of the sleep-wake cycle and other circadian rhythms. Discovered in 1958 by Aaron Lerner and colleagues, melatonin is synthesized from serotonin, a neurotransmitter derived from the amino acid tryptophan, through a series of enzymatic reactions.^[3] Its production is tightly controlled by the SCN, which relays signals from the retina to the pineal gland in response to light exposure. Light inhibits melatonin synthesis, while darkness stimulates its production, increasing melatonin levels at night. Melatonin exerts its physiological effects through two high-affinity G-protein-coupled receptors, MT₁ and MT₂, located in the SCN and other peripheral tissues. These receptors regulate circadian rhythms, promote sleep onset, and modulate various biological functions such as thermoregulation and immune responses. Beyond its role in sleep regulation, melatonin has been shown to possess antioxidant, anti-inflammatory, and immunomodulatory properties, suggesting its broader significance in health maintenance.^[4] In recent years, exogenous melatonin has gained attention as a therapeutic agent for managing sleep disorders, particularly circadian rhythm sleep-wake disorders (CRSWDs), insomnia, and conditions associated with altered sleep patterns, such as shift work and jet lag. Despite its widespread use, there is ongoing debate regarding its efficacy, optimal dosing, and safety profile, underscoring the need for continued research in this area. The primary objective of this review is to provide a comprehensive overview of melatonin's physiological role in sleep regulation by elucidating its synthesis, secretion, and regulation by the circadian system, exploring its modulation of the sleep-wake cycle, examining the clinical applications and safety of exogenous melatonin in sleep disorders, highlighting its broader physiological roles, and identifying research gaps to guide future studies and therapeutic applications.

Melatonin: Synthesis and Secretion—Biochemical Pathway

Synthesis of melatonin from serotonin in the pineal gland

Melatonin, a pivotal hormone in the regulation of circadian rhythms and sleep-wake cycles, is synthesized primarily in the pineal gland. This small, pea-shaped structure located in the epithalamus converts serotonin, an indoleamine neurotransmitter, into melatonin through a series of enzymatic reactions. The synthesis of melatonin begins with the precursor molecule, tryptophan. This essential amino acid undergoes hydroxylation via the enzyme tryptophan hydroxylase to form 5-hydroxytryptophan. Subsequently, aromatic L-amino acid

decarboxylase converts 5-hydroxytryptophan into serotonin. Serotonin serves as the substrate for melatonin synthesis, undergoing acetylation and methylation to produce melatonin.^[5]

Two critical enzymes regulate this transformation:

1. Arylalkylamine *N*-acetyltransferase (AANAT): This enzyme catalyzes the acetylation of serotonin, producing *N*-acetylserotonin. AANAT is often referred to as the “rate-limiting enzyme” in melatonin synthesis because its activity directly determines the rate of melatonin production.^[6]
2. Hydroxyindole-O-methyltransferase (HIOMT), also known as acetylserotonin-O-methyltransferase (ASMT): This enzyme catalyzes the methylation of *N*-acetylserotonin to form melatonin. The availability and activity of HIOMT influence the final step of melatonin synthesis.^[7] These enzymatic reactions predominantly occur during the dark phase, as melatonin synthesis is tightly linked to the light-dark cycle.^[8]

Enzymatic processes involved

The activity of AANAT and HIOMT is subject to intricate regulatory mechanisms, ensuring that melatonin synthesis aligns with the organism's circadian rhythm. AANAT activity is modulated by cyclic AMP (cAMP) levels, which increase during the night due to the activation of β -adrenergic receptors on pinealocytes by norepinephrine. This adrenergic stimulation enhances AANAT gene transcription and enzyme activity, thereby increasing melatonin production.^[9] HIOMT activity, while less studied, is also regulated by circadian cues and the availability of its substrate, *N*-acetylserotonin. The coordinated action of these enzymes ensures melatonin production during the night, with levels peaking in the early hours of the morning and diminishing with the onset of daylight.^[10]

Circadian regulation: Influence of the suprachiasmatic nucleus (SCN) on melatonin synthesis

The suprachiasmatic nucleus (SCN) of the hypothalamus serves as the central pacemaker of the circadian rhythm, orchestrating various physiological processes, including melatonin synthesis. The SCN receives direct photic input from the retina via the retinohypothalamic tract (RHT), which conveys information about environmental light conditions. In darkness, the SCN activates the sympathetic nervous system, leading to the release of norepinephrine from nerve terminals in the pineal gland. Norepinephrine binds to β -adrenergic receptors on pinealocytes, triggering a cascade of intracellular signaling events. This includes the activation of adenylate cyclase, increased cAMP levels, and subsequent activation of protein kinase A (PKA). PKA phosphorylates and activates AANAT, thereby enhancing melatonin synthesis. During daylight, light signals inhibit SCN activity, reducing sympathetic outflow to the pineal gland. This suppression halts norepinephrine release, leading to a decrease in cAMP levels and a subsequent decline in AANAT activity, effectively shutting down melatonin production.^[11]

Impact of light exposure on melatonin secretion

Light exposure is the most potent regulator of melatonin secretion. The retinal photoreceptors, specifically the intrinsically photosensitive retinal ganglion cells (ipRGCs), detect ambient light and relay this information to the SCN through the RHT. The photopigment melanopsin within ipRGCs is highly sensitive to blue light (460–480 nm), making this wavelength particularly effective in suppressing melatonin production.^[12] When light is perceived, the SCN inhibits the activity of the pineal gland by suppressing sympathetic output. This inhibition prevents the nocturnal rise in melatonin levels, aligning the organism's internal clock with the external light-dark cycle. Conversely, light exposure suppresses melatonin secretion via SCN-mediated inhibition, promoting wakefulness. This intricate relationship between melatonin and the circadian rhythm underscores its role as a “chronobiotic,” facilitating adaptation to varying light-dark environments. Chronic or inappropriate light exposure, such as during night shifts or jet lag, can disrupt this regulation, leading to altered melatonin rhythms and associated sleep disturbances.^[13]

The role of light in regulating melatonin has clinical implications, as appropriately timed light exposure or avoidance can help manage circadian rhythm disorders. For instance, bright light therapy is an effective intervention for conditions such as delayed sleep phase syndrome (DSPS) and seasonal affective disorder (SAD).^[14]

Physiological role of melatonin in sleep

Effects of melatonin levels on sleep onset and maintenance are mediated by its ability to promote drowsiness and enhance sleep quality. Elevated nocturnal melatonin levels reduce core body temperature, a physiological change associated with sleep initiation. Studies have shown that exogenous melatonin can significantly decrease sleep latency, particularly in individuals with delayed sleep phase syndrome or jet lag.^[15] Moreover, melatonin's influence extends to the maintenance of sleep architecture. While it does not directly induce deep or REM sleep stages, it stabilizes sleep patterns by reinforcing the circadian rhythm. Reduced melatonin secretion, often observed in aging or shift work, has been linked to fragmented sleep and increased nocturnal awakenings, highlighting its essential role in sleep.^[16]

Receptor-Mediated Mechanisms

Role of MT1 and MT2 receptors in sleep physiology

Melatonin exerts its physiological effects through high-affinity G protein-coupled receptors, MT1 and MT2, distributed across various brain regions and peripheral tissues. These receptors mediate distinct yet complementary roles in regulating sleep and circadian rhythms. The MT1 receptor primarily influences the onset of sleep by reducing neuronal excitability within the SCN. Activation of MT1 receptors during nighttime attenuates SCN activity, signaling the body to initiate sleep. In contrast, MT2 receptors regulate the phase-shifting effects of melatonin, aligning the sleep-wake cycle with environmental cues. This

dual receptor action underscores melatonin's multifaceted role in sleep physiology.^[17]

Sites of action: SCN and other brain regions

The SCN serves as the primary site of melatonin action, orchestrating the synchronization of peripheral clocks with the central circadian rhythm. MT1 and MT2 receptor activation in the SCN modulates its rhythmic activity, optimizing the timing of sleep and wakefulness. Beyond the SCN, melatonin receptors are also expressed in other brain regions, including the thalamus, hippocampus, and brainstem, where they contribute to noncircadian functions. For instance, melatonin signaling in the thalamus enhances sensory gating during sleep, while its action in the hippocampus may influence memory consolidation during the night. These widespread effects highlight the broader implications of melatonin in sleep regulation.^[18]

Chronobiotic Effects

Influence on phase shifting of circadian rhythms

Melatonin's chronobiotic properties enable it to phase-shift circadian rhythms, a mechanism particularly beneficial for individuals experiencing circadian misalignment. Administration of exogenous melatonin can advance or delay the timing of the sleep-wake cycle depending on the timing of its administration. When taken in the evening, melatonin induces a phase advance, promoting earlier sleep onset. Conversely, morning administration results in a phase delay, aligning the sleep-wake cycle with desired schedules. This phase-shifting effect is critical for managing conditions such as delayed sleep phase syndrome and jet lag. Studies demonstrate that melatonin's ability to reset the circadian clock facilitates faster adaptation to new time zones or altered work schedules.^[14]

Impact on the timing and duration of sleep

By fine-tuning the circadian clock, melatonin influences both the timing and duration of sleep. It aligns the internal biological night with the external environment, promoting efficient and restorative sleep. Research indicates that melatonin supplementation enhances total sleep time and reduces sleep onset latency in individuals with insomnia or circadian rhythm disorders. Additionally, melatonin's ability to sustain higher nocturnal levels contributes to prolonged sleep duration. This effect is particularly evident in older adults, who often experience reduced endogenous melatonin production. Supplementation in this population has been shown to improve sleep efficiency and overall quality.^[19]

Melatonin and Disorders of Sleep

Circadian Rhythm Sleep-Wake Disorders (CRSWDs) arise from misalignment between an individual's internal circadian clock and external environmental cues. Melatonin therapy targets this misalignment, aiding in the restoration of proper sleep timing.

Delayed Sleep Phase Syndrome (DSPS) and Melatonin Therapy are characterized by a significant delay in the timing of sleep onset and wakefulness, often leading to difficulty in adhering to conventional schedules. Individuals with DSPS have a normal sleep duration when allowed to follow their natural schedule, but struggle with early-morning obligations. Melatonin has demonstrated efficacy in DSPS by advancing the timing of the sleep-wake cycle. Administered in the early evening, melatonin shifts the phase of the circadian rhythm by influencing the suprachiasmatic nucleus (SCN), the brain's master circadian clock. Studies have shown that low-dose melatonin, taken 1–2 hours before the desired bedtime, can advance sleep onset and improve sleep quality in individuals with DSPS. Gradual phase shifts combined with behavioral modifications, such as light exposure in the morning, further enhance therapeutic outcomes.^[20]

Non-24-Hour Sleep-Wake Disorder primarily affects blind individuals who lack light perception, resulting in the inability to synchronize their internal circadian rhythm with the 24-hour day. This leads to a progressive delay in sleep and wake times, causing significant social and occupational disruptions. Melatonin therapy has been particularly beneficial in Non-24. Exogenous melatonin helps entrain the circadian clock by mimicking the light-dark cycle that blind individuals cannot perceive. A double-blind, placebo-controlled trial revealed that melatonin administration improved sleep onset, duration, and consolidation in patients with Non-24. Tasimelteon, a melatonin receptor agonist, has been approved for treating Non-24, offering targeted benefits for this population.^[21]

Insomnia, characterized by difficulty initiating or maintaining sleep, is classified as primary or secondary, depending on whether it occurs independently or as a result of underlying conditions such as anxiety or medical disorders. Melatonin has shown promise in managing both types.

Melatonin in Primary Insomnia is a chronic condition with no identifiable cause. Exogenous melatonin supplementation can shorten sleep latency, enhance sleep efficiency, and improve subjective sleep quality in individuals with primary insomnia. Studies have demonstrated that prolonged-release melatonin, specifically designed to mimic physiological secretion patterns, is effective in older adults with insomnia, whose endogenous melatonin production declines with age.^[22]

Melatonin in Secondary Insomnia is often associated with conditions such as depression, chronic pain, or substance use disorders. By promoting sleep initiation and reducing nighttime awakenings, melatonin complements the management of underlying conditions. Importantly, its safety profile makes it a preferred option compared to sedative-hypnotics, particularly in patients with comorbidities or elderly individuals prone to drug interactions.^[23]

Shift work and jet lag

Shift workers often face irregular sleep patterns, leading to insomnia and excessive sleepiness. Melatonin taken before daytime sleep can improve sleep quality and duration. It also enhances alertness during wake periods by stabilizing circadian rhythms. When combined with timed light exposure, melatonin helps counteract the negative effects of shift work.^[24]

Jet lag results from rapid travel across time zones, causing misalignment between the internal circadian clock and local time. Symptoms include insomnia, fatigue, and gastrointestinal issues. Melatonin helps reduce jet lag severity and duration by promoting circadian adjustment. A meta-analysis showed that melatonin, when timed to the new time zone, significantly improved sleep and reduced symptoms, especially after eastward travel.^[25]

Exogenous Melatonin: Therapeutic and Clinical Implications

Pharmacokinetics and dosage

Exogenous melatonin is rapidly absorbed in the gastrointestinal tract after oral administration, with peak plasma levels occurring within 20–60 minutes. Its bioavailability varies due to first-pass metabolism in the liver, and its half-life is approximately 30–60 minutes. Melatonin is metabolized in the liver primarily by CYP1A2 into 6-hydroxy melatonin, which is excreted in urine as sulfate and glucuronide conjugates. Optimal dosing depends on the desired therapeutic effect. Low doses (0.3–1 mg) are effective for resetting circadian rhythms, while higher doses (3–5 mg) may promote sleep onset and maintenance. For best results, melatonin should be taken 1–2 hours before bedtime to align with the natural evening rise of endogenous melatonin.^[26]

Efficacy in Sleep Disorders: Exogenous melatonin has shown efficacy in treating a variety of sleep disorders. In circadian rhythm disorders, such as delayed sleep phase syndrome, it advances sleep onset and improves sleep duration. Similarly, for jet lag and shift work sleep disorders, melatonin reduces sleep latency and enhances sleep quality when administered at the correct phase of the circadian cycle. Melatonin has been found to decrease sleep latency and modestly increase total sleep time in primary and secondary insomnia. Clinical trials have shown it to be particularly beneficial in populations with reduced endogenous melatonin levels, such as the elderly or individuals with neurodevelopmental disorders.^[23]

Safety and Adverse Effects: Melatonin is well-tolerated, with minimal short-term side effects such as headache, dizziness, and nausea. Unlike other sleep aids, melatonin does not cause dependency or withdrawal symptoms. Long-term use is considered safe, but further studies are needed to establish its safety in chronic administration. Special populations require careful consideration. In children, melatonin has been used effectively for sleep disturbances associated with autism

spectrum disorder (ASD) and attention-deficit hyperactivity disorder (ADHD), but long-term safety data are limited. For elderly individuals, melatonin supplementation compensates for diminished endogenous production, while in pregnant or lactating women, its use remains controversial due to insufficient safety data.^[27]

Nonsleep Functions of Melatonin

Antioxidant properties and neuroprotection

Melatonin is a potent antioxidant with direct and indirect effects on oxidative stress. It scavenges reactive oxygen and nitrogen species (ROS/RNS) while enhancing the activity of endogenous antioxidant enzymes like superoxide dismutase (SOD) and glutathione peroxidase (GPx). Its amphiphilic nature allows it to permeate cellular membranes, providing protection in both lipid and aqueous environments. These properties make melatonin effective in reducing oxidative damage associated with neurodegenerative disorders such as Alzheimer's and Parkinson's diseases, as well as ischemia-reperfusion injury in stroke models.^[28]

Role in immune modulation

Melatonin influences immune function through its interactions with both innate and adaptive immunity. It enhances the production of cytokines like interleukin-2 (IL-2) and interferon-gamma (IFN- γ), which bolster T-cell activity. Additionally, melatonin modulates the inflammatory response by inhibiting proinflammatory mediators such as tumor necrosis factor-alpha (TNF- α) and interleukin-6 (IL-6). These immunomodulatory effects make it a potential therapeutic agent for autoimmune conditions, sepsis, and viral infections.^[29]

Effects on other circadian-regulated physiological functions

Melatonin's circadian regulation influences metabolism, cardiovascular function, and endocrine signaling. It aids glucose homeostasis by modulating insulin secretion and sensitivity, offers cardioprotection by lowering blood pressure and preventing endothelial dysfunction, and plays a role in reproductive physiology, including seasonal fertility regulation in some species. Its multifunctionality underscores its importance in maintaining physiological homeostasis and therapeutic potential beyond sleep.^[30]

Challenges and future directions in melatonin research and sleep management

Despite progress in melatonin research, challenges remain in understanding its effects across diverse sleep disorders. While effective for jet lag and delayed sleep phase syndrome, its efficacy in primary insomnia and obstructive sleep apnea is still unclear. Variability in individual responses underscores the need to explore genetic, epigenetic, and environmental influences on melatonin metabolism and receptor sensitivity.^[31] Melatonin analogs like ramelteon and tasimelteon show potential for treating

circadian rhythm disorders in shift workers and blind individuals, though long-term safety and efficacy require further study.^[32] Personalized medicine—integrating genomics and circadian profiling—may optimize melatonin therapy, with receptor gene polymorphisms aiding in predicting outcomes and minimizing side effects.^[33] Future research should prioritize large-scale trials and interdisciplinary strategies to advance personalized, evidence-based sleep care.

Conclusion

Melatonin regulates the sleep-wake cycle and circadian rhythms, with established therapeutic use in conditions like jet lag and delayed sleep phase syndrome. While its role in other sleep disorders is still being explored, newer melatonin analogs with better receptor selectivity offer promise for specific conditions such as non-24-hour sleep-wake disorder. Personalized approaches, considering genetic and environmental factors, represent the future of melatonin therapy, with ongoing research needed to expand its clinical utility.

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Conflicts of interest

There are no conflicts of interest.

References

1. Diekelmann S, Born J. The memory function of sleep. *Nat Rev Neurosci* 2010;11:114-26.
2. Saper CB, Scammell TE, Lu J. Hypothalamic regulation of sleep and circadian rhythms. *Nature* 2005;437:1257-63.
3. Lerner AB, Case JD, Takahashi Y. Isolation of melatonin and 5-methoxyindole-3-acetic acid from bovine pineal glands. *J Biol Chem* 1960;235:1992-7.
4. Reiter RJ, Tan DX, Galano A. Melatonin: Exceeding expectations. *Physiology (Bethesda)* 2014;29:325-33.
5. Cajochen C, Kräuchi K, Wirz-Justice A. Role of melatonin in the regulation of human circadian rhythms and sleep. *J Neuroendocrinol* 2003;15:432-7.
6. Axelrod J. The pineal gland: A neurochemical transducer. *Science* 1974;184:1341-8.
7. Klein DC. Arylalkylamine N-acetyltransferase: "The Timezyme". *J Biol Chem* 2007;282:4233-7.
8. Claustrat B, Brun J, Chazot G. The basic physiology and pathophysiology of melatonin. *Sleep Med Rev* 2005;9: 11-24.
9. Reiter RJ. Melatonin synthesis: Multiplicity of regulation. In: Schwarcz R, Young SN, Brown RR, editors. *Kynurenine and Serotonin Pathways. Advances in Experimental Medicine and Biology*, vol 294. Boston, MA: Springer; 1991. https://doi.org/10.1007/978-1-4684-5952-4_13.
10. Arendt J. Melatonin and the pineal gland: Influence on mammalian seasonal and circadian physiology. *Rev Reprod* 1998;3:13-22.
11. Golombek DA, Rosenstein RE. Physiology of circadian entrainment. *Physiol Rev* 2010;90:1063-102.

12. Brainard GC, Hanifin JP, Greeson JM, Byrne B, Glickman G, Gerner E, *et al.* Action spectrum for melatonin regulation in humans: Evidence for a novel circadian photoreceptor. *J Neurosci* 2001;21:6405-12.
13. Gooley JJ, Chamberlain K, Smith KA, Khalsa SB, Rajaratnam SM, Van Reen E, *et al.* Exposure to room light before bedtime suppresses melatonin onset and shortens melatonin duration in humans. *J Clin Endocrinol Metab* 2011;96:E463-72.
14. Lewy AJ, Wehr TA, Goodwin FK, Newsome DA, Markey SP. Light suppresses melatonin secretion in humans. *Science* 1980;210:1267-9.
15. Zhdanova IV, Wurtman RJ, Regan MM, Taylor JA, Shi JP, Leclair OU. Melatonin treatment for age-related insomnia. *J Clin Endocrinol Metab* 2001;86:4727-30.
16. Pandi-Perumal SR, Trakht I, Srinivasan V, Spence DW, Maestroni GJ, Zisapel N, *et al.* Physiological effects of melatonin: Role of melatonin receptors and signal transduction pathways. *Prog Neurobiol* 2008;85:335-53.
17. Dubocovich ML, Rivera-Bermudez MA, Gerdin MJ, Masana MI. Molecular pharmacology, regulation and function of mammalian melatonin receptors. *Front Biosci* 2003;8:d1093-108.
18. Reiter RJ, Tan DX, Fuentes-Broto L, Maldonado MD. Melatonin: A multitasking molecule. *Progress Brain Res* 2014;199:127-51.
19. Brzezinski A, Vangel MG, Wurtman RJ, Norrie G, Zhdanova I, Ben-Shushan A, *et al.* Effects of exogenous melatonin on sleep: A meta-analysis. *Sleep Med Rev* 2005;9:41-50.
20. Micic G, Lovato N, Gradisar M, Ferguson SA, Burgess HJ, Lack LC. The etiology of delayed sleep phase disorder. *Sleep Med Rev* 2016;27:29-38.
21. Lockley SW, Dressman MA, Licamele L, Xiao C, Fisher DM, Flynn-Evans EE, *et al.* Tasimelteon for non-24-hour sleep-wake disorder in totally blind people (SET and RESET): Two multicentre, randomised, double-masked, placebo-controlled phase 3 trials. *Lancet* 2015;386:1754-64.
22. Lemoine P, Nir T, Laudon M, Zisapel N. Prolonged-release melatonin improves sleep quality and morning alertness in insomnia patients aged 55 years and older and has no withdrawal effects. *J Sleep Res* 2007;16:372-80.
23. Ferracioli-Oda E, Qawasmi A, Bloch MH. Meta-analysis: Melatonin for the treatment of primary sleep disorders. *PLoS One* 2013;8:e63773.
24. Burgess HJ, Revell VL, Eastman CI. A three pulse phase response curve to three milligrams of melatonin in humans. *J Physiol* 2008;586:639-47.
25. Herxheimer A, Petrie KJ. Melatonin for the prevention and treatment of jet lag. *Cochrane Database Syst Rev* 2002;CD001520.
26. Arendt J. Melatonin: Characteristics, concerns, and prospects. *J Biol Rhythms* 2005;20:291-303.
27. Pandi-Perumal SR, Zisapel N, Srinivasan V, Cardinali DP. Melatonin and sleep in aging population. *Exp Gerontol* 2005;40:911-25.
28. Reiter RJ, Mayo JC, Tan DX, Sainz RM, Alatorre-Jimenez M, Qin L. Melatonin as an antioxidant: Under promises but over delivers. *J Pineal Res* 2016;61:253-78.
29. Carrillo-Vico A, Guerrero JM, Lardone PJ, Reiter RJ. A review of the multiple actions of melatonin on the immune system. *Endocrine* 2005;27:189-200.
30. Hardeland R. Melatonin, hormone of darkness and more: Occurrence, control mechanisms, actions and bioactive metabolites. *Cell Mol Life Sci* 2008;65:2001-18.
31. Sack RL, Auckley D, Auger RR, Carskadon MA, Wright KP Jr, Vitiello MV, *et al.* American Academy of Sleep Medicine. Circadian rhythm sleep disorders: Part I, basic principles, shift work and jet lag disorders. An American Academy of Sleep Medicine review. *Sleep* 2007;30:1460-83.
32. Hardeland R. New approaches in the management of insomnia: Weighing the advantages of prolonged-release melatonin and synthetic melatoninergic agonists. *Neuropsychiatr Dis Treat* 2009;5:341-54.
33. McHill AW, Wright KP Jr. Role of sleep and circadian disruption on energy expenditure and in metabolic predisposition to human obesity and metabolic disease. *Obes Rev* 2017;18 Suppl 1:15-24.